

Ex.No: 4	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION
Date:	

AIM:

ALGORITHM:

231401116

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("indexData.csv")

print(df.head())
```

```
print(df.shape)
```

```
print(df.dtypes)
```

```
high_close = df[df["Close"] > df["Close"].mean()]

print(high_close)
```

```
high_volume = df[df["Volume"] > df["Volume"].mean()]

print(high_volume)
```

```
sorted_df = df.sort_values(
    by="Close",
    ascending=False
)

print(sorted_df)
```

```
print("Mean =", df["Close"].mean())
print("Median =", df["Close"].median())
print("Mode =", df["Close"].mode().values)
print("Standard Deviation =", df["Close"].std())
```

```
print(df.describe())
```

```
df["Date"] = pd.to_datetime(df["Date"])
```

```
df["Year"] = df["Date"].dt.year
year_avg = df.groupby("Year")["Close"].mean()

print("\nYear-wise Average Closing Price:")
print(year_avg)
```

```
corr = df[
    ["Open", "High", "Low", "Close", "Adj Close", "Volume"]
].corr()

print("\nCorrelation Matrix:")
print(corr)
```

```
plt.figure(figsize=(10, 5))

plt.plot(df["Date"], df["Close"])

plt.title("Stock Closing Price Analysis")
plt.xlabel("Date")
plt.ylabel("Close Price")

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(8, 5))

plt.bar(year_avg.index, year_avg.values)

plt.title("Year-wise Average Closing Price")
plt.xlabel("Year")
plt.ylabel("Average Close Price")

plt.show()
```

OUTPUT:

	Index	Date	Open	High	Low	Close	\
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	

	Adj Close	Volume
0	528.690002	0.0
1	527.210022	0.0
2	527.840027	0.0
3	531.119995	0.0
4	532.070007	0.0

(112457, 8)

```

Index      object
Date       object
Open       float64
High       float64
Low        float64
Close      float64
Adj Close  float64
Volume     float64
dtype: object

```

```

      Index      Date      Open      High      Low \
9991     NYA  2005-09-09  7594.640137  7667.640137  7594.640137
10042     NYA  2005-11-21  7634.580078  7679.680176  7633.250000
10043     NYA  2005-11-22  7676.640137  7716.830078  7644.040039
10044     NYA  2005-11-23  7715.359863  7768.029785  7707.919922
10045     NYA  2005-11-25  7743.359863  7754.890137  7733.100098
...      ...      ...      ...      ...      ...
106945  N225  2021-05-28  28912.539060  29194.109380  28899.660160
106946  N225  2021-05-31  29019.449220  29147.710940  28791.599610
106947  N225  2021-06-01  28998.650390  29075.470700  28611.250000
106948  N225  2021-06-02  28730.810550  29003.550780  28565.830080
106949  N225  2021-06-03  28890.390630  29157.160160  28879.150390

      Close      Adj Close      Volume
9991    7663.819824    7663.819824  1.992560e+09
10042    7676.640137    7676.640137  2.117350e+09
10043    7715.350098    7715.350098  2.291420e+09
10044    7743.339844    7743.339844  1.985400e+09
10045    7747.520020    7747.520020  7.249400e+08
...      ...      ...      ...
106945  29149.410160  29149.410160  8.060000e+07
106946  28860.080080  28860.080080  5.380000e+07
106947  28814.339840  28814.339840  4.740000e+07
106948  28946.140630  28946.140630  7.100000e+07
106949  29058.109380  29058.109380  0.000000e+00

```

```
[40632 rows x 8 columns]
```

	Index	Date	Open	High	Low \
8815	NYA	2001-01-03	6968.830078	6968.830078	6968.830078
8816	NYA	2001-01-04	6894.810059	6894.810059	6894.810059
8817	NYA	2001-01-05	6774.060059	6774.060059	6774.060059
8820	NYA	2001-01-10	6802.399902	6802.399902	6802.399902
8821	NYA	2001-01-11	6835.490234	6835.490234	6835.490234
...	...	...	...	...	...
52122	GSPTSE	2021-05-25	19621.300780	19679.900390	19553.800780
52123	GSPTSE	2021-05-26	19654.199220	19763.699220	19634.599610
52124	GSPTSE	2021-05-27	19768.099610	19835.300780	19752.500000
52125	GSPTSE	2021-05-28	19876.500000	19904.800780	19841.000000
52126	GSPTSE	2021-05-31	19905.800780	19909.599610	19705.800780

	Close	Adj Close	Volume
8815	6968.830078	6968.830078	1.880700e+09
8816	6894.810059	6894.810059	2.169400e+09
8817	6774.060059	6774.060059	1.430800e+09
8820	6802.399902	6802.399902	1.296500e+09
8821	6835.490234	6835.490234	1.411200e+09
...	...	...	...
52122	19564.099610	19564.099610	2.321558e+10
52123	19745.500000	19745.500000	2.281863e+10
52124	19774.400390	19774.400390	4.650791e+10
52125	19852.199220	19852.199220	2.270018e+10
52126	19731.000000	19731.000000	1.022570e+10

[18926 rows x 8 columns]

	Index	Date	Open	High	Low \
92396	J203.JO	2021-03-11	68516.99219	68924.99219	68516.99219
92419	J203.JO	2021-04-16	67823.09375	68741.82813	67823.09375
92394	J203.JO	2021-03-09	68426.17188	69021.59375	68051.92188
92433	J203.JO	2021-05-07	67608.89063	68664.35938	67608.89063
92395	J203.JO	2021-03-10	68655.96094	68655.96094	67941.45313
...	...	...	...	...	...
108316	N100	2005-03-28	NaN	NaN	NaN
108511	N100	2005-12-26	NaN	NaN	NaN
110051	N100	2012-01-02	NaN	NaN	NaN
110643	N100	2014-05-01	NaN	NaN	NaN
112090	N100	2019-12-25	NaN	NaN	NaN

	Close	Adj Close	Volume
92396	68775.06250	68775.06250	0.0
92419	68698.78125	68698.78125	0.0
92394	68655.96094	68655.96094	0.0
92433	68519.53125	68519.53125	0.0
92395	68516.99219	68516.99219	0.0
...	...	...	...
108316	NaN	NaN	NaN
108511	NaN	NaN	NaN
110051	NaN	NaN	NaN
110643	NaN	NaN	NaN
112090	NaN	NaN	NaN

[112457 rows x 8 columns]

Statistical Analysis:

Mean = 7657.545871842671

Median = 5194.75

Mode = [3187.620117]

Standard Deviation = 9011.510443530464

Descriptive Statistics:

	Open	High	Low	Close \
count	110253.000000	110253.000000	110253.000000	110253.000000
mean	7658.515222	7704.372961	7608.000422	7657.545872
std	9011.478913	9066.638548	8954.506981	9011.510444
min	54.869999	54.869999	54.869999	54.869999
25%	1855.030029	1864.510010	1843.979980	1855.060059
50%	5194.089844	5226.279785	5154.049805	5194.750000
75%	10134.299810	10207.820310	10060.349610	10134.830080
max	68775.062500	69403.750000	68516.992190	68775.062500

	Adj Close	Volume
count	110253.000000	1.102530e+05
mean	7657.351729	1.273975e+09
std	9011.608900	4.315783e+09
min	54.869999	0.000000e+00
25%	1854.179565	0.000000e+00
50%	5194.750000	4.329000e+05
75%	10134.830080	1.734314e+08
max	68775.062500	9.440374e+10

Year-wise Average Closing Price:

```

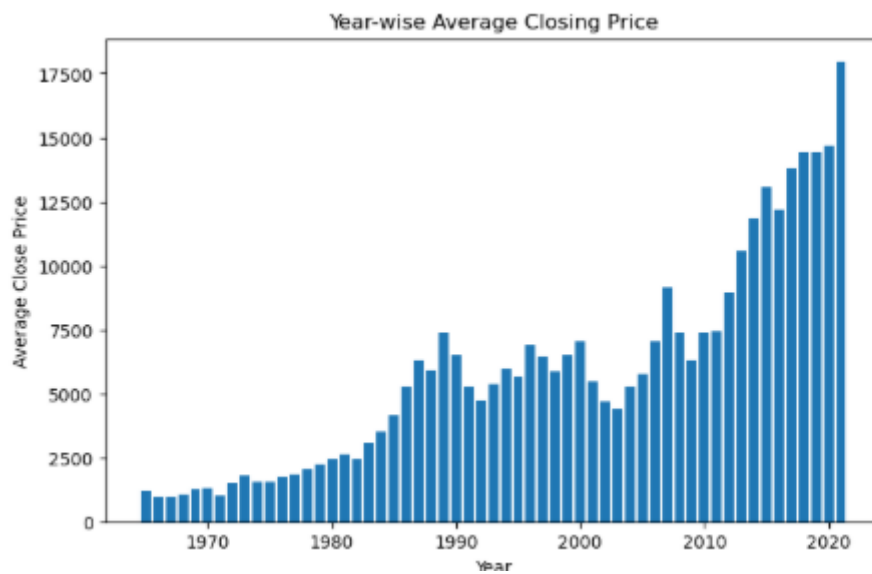
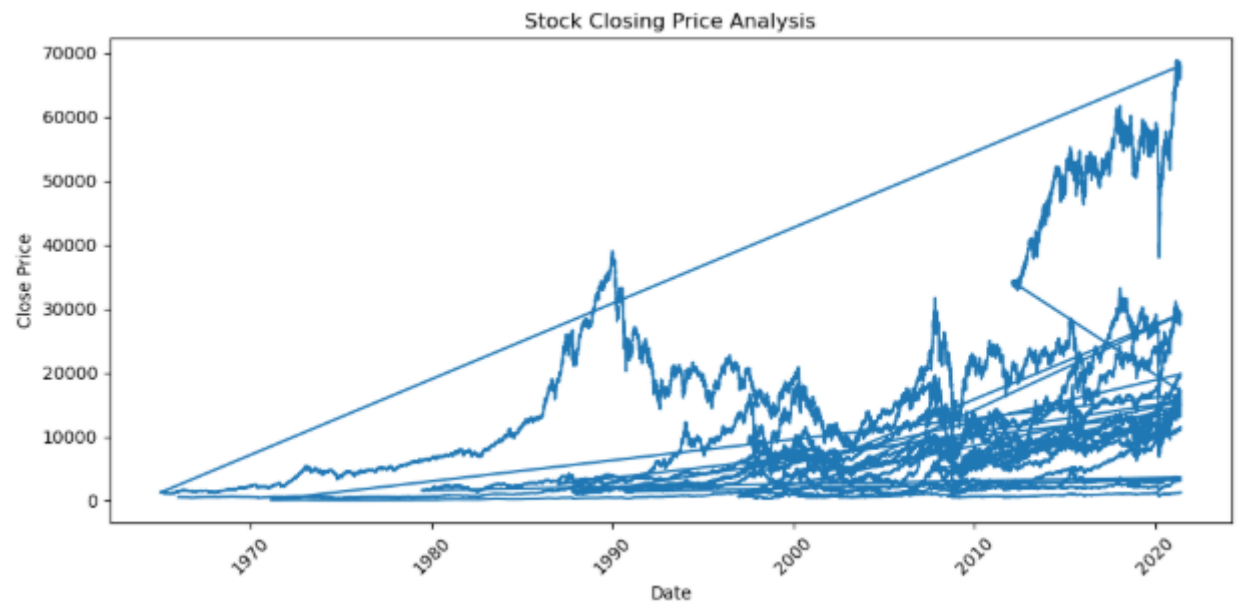
Year
1965    1198.702540
1966     976.632736
1967     971.878128
1968    1085.745443
1969    1259.602815
1970    1327.321990
1971    1036.151916
1972    1499.284429
1973    1797.495252
1974    1577.268202
1975    1572.767415
1976    1742.621396
1977    1869.089782
1978    2052.904142
1979    2221.203249
1980    2448.913243
1981    2652.156233
1982    2453.478936
1983    3076.390914
1984    3526.265449
1985    4129.361756
1986    5256.403721
1987    6307.454953
1988    5917.496005
1989    7365.583893
1990    6477.058603
1991    5296.178262
1992    4744.657594
1993    5365.075592
1994    5953.887705
1995    5653.303509
1996    6892.372371
1997    6445.372401
1998    5853.695730
1999    6496.657128
2000    7047.397209
2001    5495.652341
2002    4676.783463
2003    4407.390007
2004    5270.220391
2005    5779.419614
2006    7038.508608
2007    9163.666344
2008    7364.891839
2009    6317.328869
2010    7366.779245
2011    7441.670843
2012    8961.600646
2013    10561.298159
2014    11838.645257
2015    13065.025106
2016    12194.813886
2017    13814.377884
2018    14425.684423
2019    14428.487043
2020    14661.180592
2021    17961.546243

```

Name: Close, dtype: float64

Correlation Matrix:

	Open	High	Low	Close	Adj Close	Volume
Open	1.000000	0.999959	0.999944	0.999907	0.999907	0.138673
High	0.999959	1.000000	0.999931	0.999955	0.999955	0.138510
Low	0.999944	0.999931	1.000000	0.999960	0.999960	0.138549
Close	0.999907	0.999955	0.999960	1.000000	1.000000	0.138555
Adj Close	0.999907	0.999955	0.999960	1.000000	1.000000	0.138560
Volume	0.138673	0.138510	0.138549	0.138555	0.138560	1.000000



RESULT: